AI Intern Case Study Assignment: Automated China–Indonesia Tariff Code Mapping Web App

Assignment Overview

Design and build a web application using Generative AI-first workflows to help users map import and export product classification codes and tariff information between China and Indonesia. The app should help a user start with a product description, HS code, or local tariff code in one country and identify the most likely corresponding classifications in the other country.

Duration: 1 week.

Core Task

Create a prototype web app that maps tariff and customs classification codes between China and Indonesia. Users should be able to search by:

- Product description.
- HS code.
- Local tariff code.

For each search, the app must return the top 5 most likely code matches for the other country, ranked by confidence or probability. Each result should include:

- Matched code.
- Product description.
- Confidence/probability score.
- Match label: exact match, likely match, partial match, or manual review required.
- Explanation of why the result was suggested.
- Tariff rate or tariff indicator where available.
- Source references.

The app must support:

- One-to-many mappings.
- Ambiguous or incomplete product descriptions.
- Differences between China and Indonesia national code extensions beyond 6 digits.
- Cases where no clean match exists and the user should be told manual review is needed.

Example workflow:

User enters a product description or tariff code → app finds the nearest HS anchor → compares likely codes in the other country → returns top 5 matches with explanations, tariffs, and confidence levels.

Allowed Tools & Constraints

- **Primary:** Generative AI tools such as Cursor, Replit, Lovable, Windsurf, ChatGPT, Claude, or similar AI-enabled platforms.
- **Research/Data:** Public tariff schedules, customs search tools, government tariff publications, trade databases, GitHub, and public classification resources.
- **No:** Fully manual coding from scratch; fabricated tariff rates; unsupported mappings; non-public customs databases unless explicitly approved.
- AI should drive most of the workflow, but the app must include a traceable logic layer that shows why a mapping was returned.
- The system should distinguish between high-confidence crosswalks and uncertain inferences.
- Do not imply legal certainty or customs binding-ruling status unless the source explicitly supports that.

Deliverables

Submit via Google Drive folder or GitHub repo:

1. **Prototype Web App**

- Live link or runnable local/web version.
- Search interface supporting product description, HS code, and local tariff code.
- Country-to-country mapping in both directions: China → Indonesia and Indonesia → China.
- Results page showing top 5 matches with confidence, explanation, tariff data, and source references.

2. **Demo Video (OPTIONAL)**

- 3–5 minutes showing:
    - Search by description.
    - Search by HS code.
    - Search by local tariff code.
    - Ambiguous-result handling.
    - Example of one-to-many mapping.
    - Example of manual review required.

3. **Build Documentation** (2–4 pages or equivalent slides)

- Tools used.
- Key prompts and prompt iterations.
- Workflow or system diagram.
- Data sources and why they were selected.
- Matching logic and ranking methodology.
- How the app handles fuzzy product descriptions.
- How the app handles national code differences beyond 6 digits.
- Limitations and unresolved classification risks.
- What AI did well versus what required human judgment.

4. **Reflection** (1 paragraph)

- What worked best.
- Where ambiguity or weak matching appeared.
- What you would improve in the next version.

Functional Requirements

Your app should include the following:

Search inputs

- Product description search.
- HS code search.
- China local tariff/customs code search.
- Indonesia local tariff/BTKI/AHTN-based code search.

Result outputs

- Top 5 candidate matches.
- Confidence score for each result.
- Match type label.
- Short explanation.
- Tariff rate, tariff type, or tariff note where available.
- Source links or citation references.

Matching behavior

- Must support one-to-many mappings.

- Must work even when the product description is incomplete or vague.
- Must identify when country-specific extensions create divergence.
- Must return "manual review required" when certainty is too low.

Explainability

- Show whether the match was based on shared HS digits, semantic description similarity, tariff-book structure, or other logic.
- Show what data source the match came from.